

MUHAMMAD TAYYAB PARACHA

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ABOUT ME

Senior Software Engineer with 5+ years of experience designing scalable backend systems, cloud-native microservices, and high-performance APIs using Python, Flask, FastAPI, AWS, PostgreSQL, MongoDB, Redis, and event-driven architecture.

Proven ability to lead engineering teams, design system architecture, and deliver secure, reliable, production-grade software. Experienced in transaction-heavy systems, real-time platforms, data-intensive applications, and developer-facing APIs with a strong focus on performance, reliability, and scalability.

TECHNICAL STRENGTHS

Languages & Frameworks	Python, Django, Flask, FastAPI, Microservices, WebSockets, REST APIs, JavaScript, React
Databases	PostgreSQL, MySQL, SQLite, MongoDB, DynamoDB, Redis
Cloud & DevOps	AWS (Lambda, EC2, ECS, S3, CloudWatch, SQS, SNS, API Gateway, CloudFormation), Azure (Virtual Machines, Functions, Blob Storage, SQL Database, Monitor, Cognitive Services, Speech Services, Translator), Docker, GitHub Actions, CI/CD
Data & AI	Pandas, NumPy, Jupyter Notebook, Scikit-learn, PyTorch, Hugging Face, OpenAI APIs, OpenCV
Testing & Automation	Playwright, Selenium, API Testing, Unit Testing, Integration Testing, Test-Driven Development (TDD)
Software Engineering	SDLC, System Design, Agile/Scrum
Project Management Tools	Jira, Monday, GitHub, GitLab, ClickUp, Trello, Asana

WORK EXPERIENCE

Devsinc, Islamabad <i>Senior Software Engineer</i>	Mar 2022 – Present
- Led a team of 3 engineers, owning backend architecture, sprint delivery, code reviews, and production quality across multiple client systems. - Designed multi-tenant backend platforms using Python, Flask/FastAPI, PostgreSQL, MongoDB, Redis, and AWS services. - Built event-driven workflows using AWS SQS, SNS, and Lambda to support asynchronous processing and high-concurrency workloads. - Improved API performance by 30%+ through query optimization, caching, indexing, and service-level refactoring. - Managed CI/CD pipelines and AWS infrastructure provisioning using CloudFormation and GitHub Actions. - Integrated third-party APIs and designed reliable retry, logging, and failure-handling mechanisms for distributed services. - Reduced AWS infrastructure cost through resource optimization, autoscaling improvements, and better service utilization.	

Turing (US - Remote)

Aug 2024 - Jan 2025

Senior Backend Engineer

- Developed backend services for a large-scale supply chain platform for a client valued at \$15B, supporting real-time operations and high-volume data workflows.
- Designed and implemented REST APIs and microservices for transactional supply chain workflows using Python, FastAPI/Django, PostgreSQL, MongoDB, and AWS.
- Optimized database schemas, indexes, and queries, improving API response times by 30–40%.
- Built cloud-native deployment infrastructure using AWS CDK, improving consistency across environments.
- Implemented monitoring, logging, and fault-tolerant processing to reduce downtime and improve production reliability.
- Collaborated with distributed engineering teams to define backend architecture standards and deployment practices.

Moaice

April 2020 – Jan 2022

Software Engineer

- Delivered 50+ client projects across backend development, automation, web scraping, AI/ML prototypes, and full-stack web applications.
- Built REST APIs, admin dashboards, scraping pipelines, and automation tools using Python, Django, Flask, FastAPI, React, Selenium, and cloud services.
- Maintained high client satisfaction and earned Top Performer recognition through consistent delivery, communication, and code quality.

PROJECTS

KUDO.ai

kudo.ai

- Designed a real-time multilingual translation platform using FastAPI and WebSockets for seamless video communication.
- Integrated Azure Speech-to-Text, Google Translation API, ChatGPT API, and Azure SDK.
- Enabled accurate real-time transcription and translation with low-latency performance.

Faerie Tales Creamery

faerietalescreamery.com

- Developed backend services for a mobile ice cream truck operations platform supporting real-time cart tracking, menus, discounts, inventory, and route management.
- Built and maintained Strapi CMS backend and FastAPI services for driver app, user app, and admin web platform.
- Implemented real-time tracking workflows, promotion management, and optimized APIs for scalability and reliability.

Window

thisiswindow.com

- Engineered a fashion aggregation platform processing data from 70+ e-commerce sites.
- Built scalable scrapers using Scrapy and Selenium on AWS Lambda.
- Designed DynamoDB schemas and developed Flask APIs to serve thousands of users efficiently.

Intellirent

myintellirent.com

- Intellirent is a SaaS platform that automates the rental leasing process, enabling property managers to market listings, screen tenants, and manage applications in one system.
- Provides automated background checks, income verification, and centralized communication tools to streamline approvals.
- Contributed to platform features improving leasing workflows and reducing manual effort.

EDUCATION

COMSATS University Islamabad

BSCS (2018 - 2022)